Sum and Difference Formulas

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Exact values with sum and difference formulas**

Use a sum or difference formula to find the exact value of $\cos 75°$.

$\cos 75° = \cos(45° + 30°) = \cos 45°\cos 30° - \sin 45°\sin 30° = \frac{\sqrt{6}-\sqrt{2}}{4}$.

2 **[Foundational]** • **Target: Exact values with sum and difference formulas**

Find the exact value of $\sin \frac{\pi}{12}$.

$\sin \frac{\pi}{12} = \sin\left(\frac{\pi}{3} - \frac{\pi}{4}\right) = \sin\frac{\pi}{3}\cos\frac{\pi}{4} - \cos\frac{\pi}{3}\sin\frac{\pi}{4} = \frac{\sqrt{6}-\sqrt{2}}{4}$.

3 **[Foundational]** • **Target: Exact values with sum and difference formulas**

Find the exact value of $\tan 105°$.

$\tan 105° = \tan(60° + 45°) = \frac{\sqrt{3}+1}{1-\sqrt{3}}$. Rationalizing: $\frac{(\sqrt{3}+1)^2}{(1-\sqrt{3})(1+\sqrt{3})} = \frac{4+2\sqrt{3}}{-2} = -2-\sqrt{3}$.

4 **[Foundational]** • **Target: Working with given values**

Angles A and B are both in Quadrant I with $\sin A = \frac{4}{5}$ and $\cos B = \frac{5}{13}$. Find:

a $\sin(A+B)$ $\cos A = \frac{3}{5}$, $\sin B = \frac{12}{13}$, so $\sin(A+B) = \frac{4}{5}\cdot\frac{5}{13} + \frac{3}{5}\cdot\frac{12}{13} = \frac{56}{65}$.

b $\cos(A+B)$ $\cos(A+B) = \frac{3}{5}\cdot\frac{5}{13} - \frac{4}{5}\cdot\frac{12}{13} = -\frac{33}{65}$.

c the quadrant of $A+B$ Sine positive, cosine negative: Quadrant II.

(a) $\cos A = \frac{3}{5}$, $\sin B = \frac{12}{13}$, so $\sin(A+B) = \frac{4}{5}\cdot\frac{5}{13} + \frac{3}{5}\cdot\frac{12}{13} = \frac{56}{65}$.; (b) $\cos(A+B) = \frac{3}{5}\cdot\frac{5}{13} - \frac{4}{5}\cdot\frac{12}{13} = -\frac{33}{65}$.; (c) Sine positive, cosine negative: Quadrant II.

5 **[Foundational]** • **Target: Working with given values**

Use a sum or difference formula to simplify:

a $\sin\left(x + \frac{\pi}{2}\right) = \cos x$

b $\cos(x - \pi) = -\cos x$

(a) $= \cos x$; (b) $= -\cos x$

6 **[Foundational]** • **Target: Working with given values**

Angles A and B are both in Quadrant II with $\sin A = \frac{8}{17}$ and $\sin B = \frac{3}{5}$. Find $\cos(A-B)$ exactly.

In QII, $\cos A = -\frac{15}{17}$ and $\cos B = -\frac{4}{5}$.
$\cos(A-B) = \cos A\cos B + \sin A\sin B = \left(-\frac{15}{17}\right)\left(-\frac{4}{5}\right) + \frac{8}{17}\cdot\frac{3}{5} = \frac{60}{85} + \frac{24}{85} = \frac{84}{85}$.

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7 [Foundational] • Target: Verifying identities using sum and difference formulas

Verify the identity $\cos(x+y)+\cos(x-y)=2\cos x\cos y$.

$\cos(x+y)=\cos x\cos y-\sin x\sin y$ and $\cos(x-y)=\cos x\cos y+\sin x\sin y$. Adding: the $\sin x\sin y$ terms cancel, leaving $2\cos x\cos y$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Verifying identities using sum and difference formulas

Verify the identity $\sin(x+y)-\sin(x-y)=2\cos x\sin y$.

$\sin(x+y)=\sin x\cos y+\cos x\sin y$ and $\sin(x-y)=\sin x\cos y-\cos x\sin y$. Subtracting: the $\sin x\cos y$ terms cancel, leaving $2\cos x\sin y$.

9 [Intermediate] • Target: Verifying identities using sum and difference formulas

Show that $\tan\left(x+\frac{\pi}{4}\right)=\frac{1+\tan x}{1-\tan x}$ using the tangent sum formula $\tan(A+B)=\frac{\tan A+\tan B}{1-\tan A\tan B}$.

With $B=\frac{\pi}{4}$, $\tan B=1$: $\tan\left(x+\frac{\pi}{4}\right)=\frac{\tan x+1}{1-\tan x\cdot 1}=\frac{1+\tan x}{1-\tan x}$.

10 [Intermediate] • Target: Applications

Two musical tones combine to form a sound wave modeled by $y=\sin\left(x+\frac{\pi}{6}\right)$. Expand this expression using the sum formula, and evaluate the coefficient of $\cos x$ exactly.

$\sin\left(x+\frac{\pi}{6}\right)=\sin x\cos\frac{\pi}{6}+\cos x\sin\frac{\pi}{6}=\frac{\sqrt{3}}{2}\sin x+\frac{1}{2}\cos x$. The coefficient of $\cos x$ is $\frac{1}{2}$.

11 [Intermediate] • Target: Additional practice

Find the exact value of $\cos\frac{\pi}{12}$.

$\cos\frac{\pi}{12}=\cos\left(\frac{\pi}{3}-\frac{\pi}{4}\right)=\cos\frac{\pi}{3}\cos\frac{\pi}{4}+\sin\frac{\pi}{3}\sin\frac{\pi}{4}=\frac{\sqrt{6}+\sqrt{2}}{4}$.

12 [Intermediate] • Target: Additional practice

Angles A and B satisfy $\tan A=\frac{3}{4}$ (QI) and $\tan B=\frac{5}{12}$ (QI). Find $\tan(A+B)$ using the tangent sum formula.

$\tan(A+B)=\frac{\tan A+\tan B}{1-\tan A\tan B}=\frac{3/4+5/12}{1-(3/4)(5/12)}=\frac{14/12}{1-15/48}=\frac{7/6}{33/48}=\frac{7}{6}\cdot\frac{48}{33}=\frac{56}{33}$.

13 [Intermediate] • Target: Additional practice

Simplify $\cos\left(\frac{\pi}{2}-x\right)\cos x+\sin\left(\frac{\pi}{2}-x\right)\sin x$ using the cosine difference formula in reverse.

This matches the pattern $\cos A\cos B+\sin A\sin B=\cos(A-B)$ with $A=\frac{\pi}{2}-x$, $B=x$: $=\cos\left(\frac{\pi}{2}-2x\right)$.

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14 [Intermediate] • Target: A worked derivation

Starting from the cosine difference formula $\cos(A-B)=\cos A\cos B+\sin A\sin B$, derive the cosine sum formula $\cos(A+B)$ by substituting $B\to -B$.

$\cos(A+B)=\cos(A-(-B))=\cos A\cos(-B)+\sin A\sin(-B)$. Since cosine is even and sine is odd: $=\cos A\cos B-\sin A\sin B$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: A worked derivation • Cross-Lesson: 8.02 Graphing Rational Functions

Find the exact value of $\sin 15°\cos 15°$ using a sum/difference-related double angle shortcut (recall $\sin(2\theta)=2\sin\theta\cos\theta$).

$\sin 15°\cos 15°=\frac{1}{2}(2\sin 15°\cos 15°)=\frac{1}{2}\sin 30°=\frac{1}{2}\cdot\frac{1}{2}=\frac{1}{4}$.

16 [Challenging] • Target: Sum and difference formulas with tables of values • Cross-Lesson: 8.03 Introduction to Limits

Given $\sin A=\frac{7}{25}$ (QI) and $\cos B=\frac{4}{5}$ (QI), find $\sin(A-B)$ exactly.

$\cos A=\frac{24}{25}$, $\sin B=\frac{3}{5}$. $\sin(A-B)=\sin A\cos B-\cos A\sin B=\frac{7}{25}\cdot\frac{4}{5}-\frac{24}{25}\cdot\frac{3}{5}=\frac{28}{125}-\frac{72}{125}=-\frac{44}{125}$.

17 [Challenging] • Target: Sum and difference formulas with tables of values • Cross-Lesson: 9.01 Arithmetic Sequences and Series

Given $\tan A=2$ (QI) and $\tan B=3$ (QI), find $\tan(A-B)$ using the tangent difference formula.

$\tan(A-B)=\frac{\tan A-\tan B}{1+\tan A\tan B}=\frac{2-3}{1+6}=-\frac{1}{7}$.

18 [Challenging] • Target: Expanding a general sinusoidal expression • Cross-Lesson: 9.02 Geometric Sequences and Series

Expand $\cos\left(x-\frac{\pi}{3}\right)$ fully using the difference formula, leaving the answer in terms of $\sin x$ and $\cos x$.

$\cos\left(x-\frac{\pi}{3}\right)=\cos x\cos\frac{\pi}{3}+\sin x\sin\frac{\pi}{3}=\frac{1}{2}\cos x+\frac{\sqrt{3}}{2}\sin x$.

19 [Challenging] • Target: A geometric interpretation • Cross-Lesson: 9.03 The Binomial Theorem

Explain, using the unit circle, why $\cos(A-B)$ (not $\cos A-\cos B$) correctly gives the cosine of the angle between two points at angles A and B on the unit circle.

The angle between two points on the unit circle at angles A and B is $|A-B|$, and by definition the cosine of that INCLUDED angle is $\cos(A-B)$ — a genuinely different quantity from the difference of the individual cosine values $\cos A-\cos B$, which has no direct geometric meaning as an angle's cosine.

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20 [Challenging] • Target: Solving equations using sum and difference formulas • Cross-Lesson: 1.01 Functions and Function Notation

Solve $\sin\left(x+\frac{\pi}{6}\right)=\frac{1}{2}$ for $x\in[0,2\pi)$ by first expanding the left side.

Expand: $\sin x\cos\frac{\pi}{6}+\cos x\sin\frac{\pi}{6}=\frac{1}{2}\Rightarrow\frac{\sqrt{3}}{2}\sin x+\frac{1}{2}\cos x=\frac{1}{2}$. Alternatively, recognizing $\sin(\theta)=\frac{1}{2}$ directly: $x+\frac{\pi}{6}=\frac{\pi}{6}$ or $x+\frac{\pi}{6}=\frac{5\pi}{6}$, giving $x=0$ or $x=\frac{2\pi}{3}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Sine Sum and Difference Identities

State the formula for $\sin(A+B)$.

a. $\sin A\cos B+\cos A\sin B$ ✓

b. $\sin A\cos B-\cos A\sin B$

c. $\cos A\cos B-\sin A\sin B$

d. $\sin A+\sin B$

The sine sum formula is $\sin(A+B)=\sin A\cos B+\cos A\sin B$.

22 [Foundational] • Target: Sine Sum and Difference Identities

Find the exact value of $\sin(75^\circ)$ using $75^\circ=45^\circ+30^\circ$.

a. $\frac{\sqrt{6}+\sqrt{2}}{4}$ ✓

b. $\frac{\sqrt{6}-\sqrt{2}}{4}$

c. $\frac{\sqrt{3}+1}{2}$

d. $\frac{\sqrt{2}+1}{2}$

$\sin(45^\circ)\cos(30^\circ)+\cos(45^\circ)\sin(30^\circ)=\left(\frac{\sqrt{2}}{2}\right)\left(\frac{\sqrt{3}}{2}\right)+\left(\frac{\sqrt{2}}{2}\right)\left(\frac{1}{2}\right)=\frac{\sqrt{6}+\sqrt{2}}{4}$.

23 [Foundational] • Target: Sine Sum and Difference Identities

Simplify $\sin(x+\pi)$.

a. $-\sin x$ ✓

b. $\sin x$

c. $-\cos x$

d. $\cos x$

$\sin x\cos\pi+\cos x\sin\pi=\sin x(-1)+\cos x(0)=-\sin x$.

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24 [Foundational] • Target: Cosine Sum and Difference Identities

State the formula for $\cos(A - B)$.

a. $\cos A\cos B + \sin A\sin B$ ✓

b. $\cos A\cos B - \sin A\sin B$

c. $\sin A\sin B - \cos A\cos B$

d. $\cos A - \cos B$

The cosine difference formula is $\cos(A - B) = \cos A\cos B + \sin A\sin B$.

25 [Foundational] • Target: Cosine Sum and Difference Identities

Find the exact value of $\cos(15^\circ)$ using $15^\circ = 45^\circ - 30^\circ$.

a. $\frac{\sqrt{6}+\sqrt{2}}{4}$ ✓

b. $\frac{\sqrt{6}-\sqrt{2}}{4}$

c. $\frac{\sqrt{3}-1}{4}$

d. $\frac{\sqrt{2}+1}{4}$

$\cos(45^\circ)\cos(30^\circ) + \sin(45^\circ)\sin(30^\circ) = \frac{\sqrt{2}}{2}\frac{\sqrt{3}}{2} + \frac{\sqrt{2}}{2}\frac{1}{2} = \frac{\sqrt{6}+\sqrt{2}}{4}$.

26 [Foundational] • Target: Cosine Sum and Difference Identities

Simplify $\cos(x - \pi/2)$.

a. $\sin x$ ✓

b. $-\sin x$

c. $\cos x$

d. $-\cos x$

$\cos x\cos(\pi/2) + \sin x\sin(\pi/2) = \cos x(0) + \sin x(1) = \sin x$.

27 [Foundational] • Target: Tangent Sum and Difference Identities

State the formula for $\tan(A + B)$.

a. $\frac{\tan A + \tan B}{1 - \tan A\tan B}$ ✓

b. $\frac{\tan A - \tan B}{1 + \tan A\tan B}$

c. $\frac{\tan A + \tan B}{1 + \tan A\tan B}$

d. $\tan A + \tan B$

The tangent sum formula is $\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A\tan B}$.

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28 [Foundational] • Target: Tangent Sum and Difference Identities

If $\tan A = 1/2$ and $\tan B = 1/3$, find $\tan(A + B)$.

a. 1 ✓

b. 5/6

c. 5/7

d. 2/3

$\frac{1/2 + 1/3}{1 - (1/2)(1/3)} = \frac{5/6}{1 - 1/6} = \frac{5/6}{5/6} = 1.$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Tangent Sum and Difference Identities

Evaluate $\frac{\tan(80^\circ) - \tan(20^\circ)}{1 + \tan(80^\circ)\tan(20^\circ)}$.

a. $\sqrt{3}$ ✓

b. $1/\sqrt{3}$

c. 1

d. 0

This matches the formula for $\tan(80^\circ - 20^\circ) = \tan(60^\circ) = \sqrt{3}$.

30 [Intermediate] • Target: Compound Angle Problems with Given Ratios

Given $\sin\alpha = 3/5$ (Quadrant I) and $\cos\beta = -5/13$ (Quadrant II), find $\sin(\alpha + \beta)$.

a. 33/65 ✓

b. −33/65

c. 63/65

d. −63/65

$\cos\alpha = 4/5$. $\sin\beta = \sqrt{1 - (-5/13)^2} = 12/13$.

$\sin(\alpha + \beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta = (3/5)(-5/13) + (4/5)(12/13) = -15/65 + 48/65 = 33/65.$

31 [Intermediate] • Target: Compound Angle Problems with Given Ratios

For the same angles ($\cos\alpha = 4/5, \sin\alpha = 3/5, \cos\beta = -5/13, \sin\beta = 12/13$), find $\cos(\alpha + \beta)$.

a. −56/65 ✓

b. 56/65

c. −16/65

d. 16/65

$\cos(\alpha + \beta) = \cos\alpha\cos\beta - \sin\alpha\sin\beta = (4/5)(-5/13) - (3/5)(12/13) = -20/65 - 36/65 = -56/65.$

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32 [Intermediate] • Target: Compound Angle Problems with Given Ratios

In which quadrant does $\alpha + \beta$ lie in the previous problems?

a. Quadrant II ✓

b. Quadrant I

c. Quadrant III

d. Quadrant IV

Since $\cos(\alpha + \beta) = -56/65 < 0$ and $\sin(\alpha + \beta) = 33/65 > 0$, the angle $\alpha + \beta$ terminates in Quadrant II.

33 [Intermediate] • Target: Tangent Sum and Difference Identities

Evaluate $\sin(45^\circ + 30^\circ)$ using a sum/difference formula. (variant 10)

a. $6 - 24$

b. 22

c. $2 - 64$

d. $6 + 24$ ✓

$(45^{+}30) = 45^{3}0^{+}45^{3}0^{=}6 + 24.$

34 [Intermediate] • Target: Compound Angle Problems with Given Ratios

Evaluate $\cos(45^\circ - 30^\circ)$ using a sum/difference formula. (variant 5)

a. $6 + 24$ ✓

b. $2 - 64$

c. 32

d. $6 - 24$

$(45^{-}30) = 45^{3}0^{+}45^{3}0^{=}6 + 24.$

35 [Intermediate] • Target: Exact values with sum and difference formulas

Evaluate $\cos(60^\circ + 45^\circ)$ using a sum/difference formula. (variant 12)

a. $6 + 24$

b. $6 - 24$

c. $\frac{1}{2}$

d. $2 - 64$ ✓

$(60^{+}45) = 60^{4}5^{-}60^{4}5^{=}2 - 64.$

36 [Intermediate] • Target: Working with given values

Evaluate $\cos(45^\circ - 30^\circ)$ using a sum/difference formula. (variant 10)

a. $2 - 64$

b. 32

c. $\frac{1}{2}$

d. $6 + 24$ ✓

$(45^{-}30) = 45^{3}0^{+}45^{3}0^{=}6 + 24.$

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37 [Intermediate] • Target: Verifying identities using sum and difference formulas

Evaluate $\sin(60^\circ - 45^\circ)$ using a sum/difference formula. (variant 7)

a. $6 + 24$

b. 22

c. $6 - 24$ ✓

d. $\frac{1}{2}$

$(60 - 45) = 60 \cdot 45 - 60 \cdot 45 = 6 - 24$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Applications • Cross-Lesson: 9.02 Geometric Sequences and Series

Evaluate $\sin(60^\circ - 45^\circ)$ using a sum/difference formula. (variant 9)

a. $2 - 64$

b. 22

c. $6 + 24$

d. $6 - 24$ ✓

$(60 - 45) = 60 \cdot 45 - 60 \cdot 45 = 6 - 24$.

39 [Challenging] • Target: Additional practice • Cross-Lesson: 9.03 The Binomial Theorem

Evaluate $\cos(45^\circ - 30^\circ)$ using a sum/difference formula. (variant 7)

a. $6 + 24$ ✓

b. $2 - 64$

c. 32

d. 22

$(45 - 30) = 45 \cdot 30 + 45 \cdot 30 = 6 + 24$.

40 [Challenging] • Target: A worked derivation • Cross-Lesson: 1.01 Functions and Function Notation

Evaluate $\sin(45^\circ + 30^\circ)$ using a sum/difference formula. (variant 3)

a. $2 - 64$

b. 32

c. 22

d. $6 + 24$ ✓

$(45 + 30) = 45 \cdot 30 + 45 \cdot 30 = 6 + 24$.

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41 [Challenging] • Target: Sum and difference formulas with tables of values • Cross-Lesson: 1.02 Domain and Range

Evaluate $\cos(60^\circ + 45^\circ)$ using a sum/difference formula.

a. $2 - 64$ ✓

b. $\frac{1}{2}$

c. 22

d. $6 - 24$

$(60^{+}45^{)} = 60^{4}5^{-}60^{4}5^{=} 2 - 64.$

42 [Challenging] • Target: Expanding a general sinusoidal expression • Cross-Lesson: 1.03 Average Rate of Change

Evaluate $\sin(45^\circ + 30^\circ)$ using a sum/difference formula. (variant 8)

a. $2 - 64$

b. $\frac{1}{2}$

c. $6 + 24$ ✓

d. $6 - 24$

$(45^{+}30^{)} = 45^{3}0^{+}45^{3}0^{=} 6 + 24.$

43 [Challenging] • Target: A geometric interpretation • Cross-Lesson: 2.01 Right-Triangle Trigonometry

Evaluate $\sin(45^\circ + 30^\circ)$ using a sum/difference formula. (variant 2)

a. $\frac{1}{2}$

b. $6 + 24$ ✓

c. $2 - 64$

d. 22

$(45^{+}30^{)} = 45^{3}0^{+}45^{3}0^{=} 6 + 24.$

44 [Challenging] • Target: Solving equations using sum and difference formulas • Cross-Lesson: 2.02 Radian Measure and Arc Length

Evaluate $\sin(60^\circ - 45^\circ)$ using a sum/difference formula. (variant 3)

a. 32

b. $\frac{1}{2}$

c. $6 + 24$

d. $6 - 24$ ✓

$(60^{-}45^{)} = 60^{4}5^{-}60^{4}5^{=} 6 - 24.$

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45 [Challenging] • Target: Sine Sum and Difference Identities • Cross-Lesson: 2.03 Graphs of Sine and Cosine Functions

Evaluate $\cos(60^\circ + 45^\circ)$ using a sum/difference formula. (variant 4)

a. $6 + 24$

b. $2 - 64$ ✓

c. $6 - 24$

d. 32

$(60^{+}45^{)} = 60^{4}5^{-}60^{4}5^{=}2 - 64.$

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